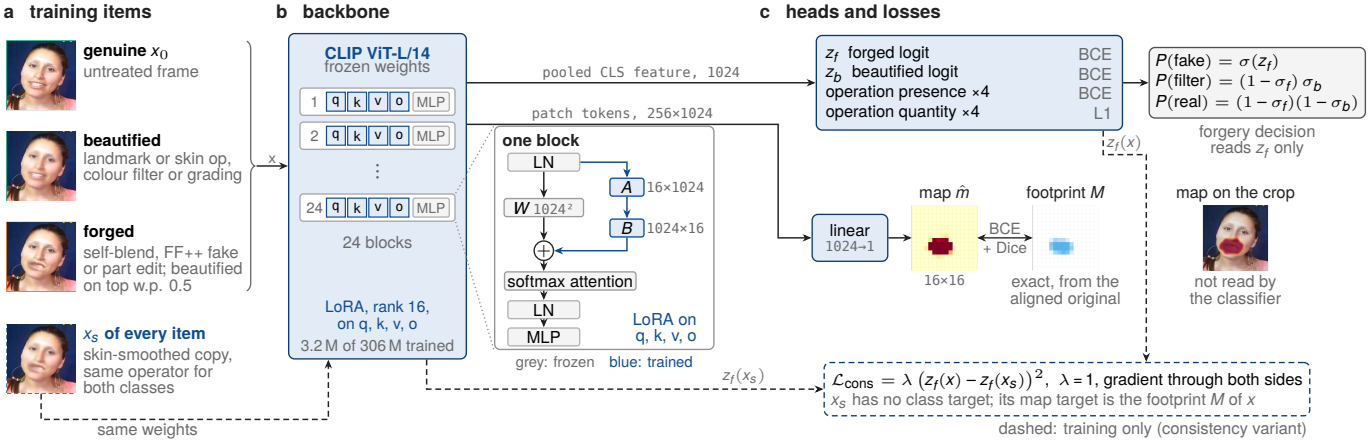

a training items
genuine x_0
untreated frame
beautified
landmark or skin op, colour filter or grading
forged
self-blend, FF++ fake or part edit; beautified on top w.p. 0.5
x_s of every item
skin-smoothed copy, same operator for both classes
same weights
x
b backbone
CLIP ViT-L/14
frozen weights
1 q k v o MLP
2 q k v o MLP
24 q k v o MLP
24 blocks
LoRA, rank 16, on q, k, v, o
3.2 M of 306 M trained
pooled CLS feature, 1024
patch tokens, 256×1024
one block
LN
A 16×1024
W 1024²
B 1024×16
softmax attention
LN
MLP
LoRA on q, k, v, o
grey: frozen blue: trained
c heads and losses
z_f forged logit BCE
z_b beautified logit BCE
operation presence ×4 BCE
operation quantity ×4 L1
$P(\text{fake}) = \sigma(z_f)$
$P(\text{filter}) = (1-\sigma_f)\,\sigma_b$
$P(\text{real}) = (1-\sigma_f)(1-\sigma_b)$
forgery decision reads z_f only
$z_f(x)$
linear 1024→1
map $\hat{m}$
16×16
BCE + Dice
footprint M
exact, from the aligned original
map on the crop
not read by the classifier
$z_f(x_s)$
$\mathcal{L}_{\text{cons}} = \lambda\,(z_f(x) - z_f(x_s))^2$, $\lambda = 1$, gradient through both sides
x_s has no class target; its map target is the footprint M of x
dashed: training only (consistency variant)